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Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – March 2025

Second Semester

Artificial Intelligence

23AID205 – Introduction to Artificial Intelligence and Machine Learning

Duration: Two hours Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Understand the foundational concepts of AI, intelligent agents, and environments, and describe the difference between AI, Machine Learning, Deep Learning, and Data Science.
CO02	Apply search algorithms and optimization techniques to solve problems using both uninformed and informed search strategies, as well as local search and optimization algorithms.

Instructions: Answer all questions. Show all relevant steps and logic according to sections and marks.

PART A

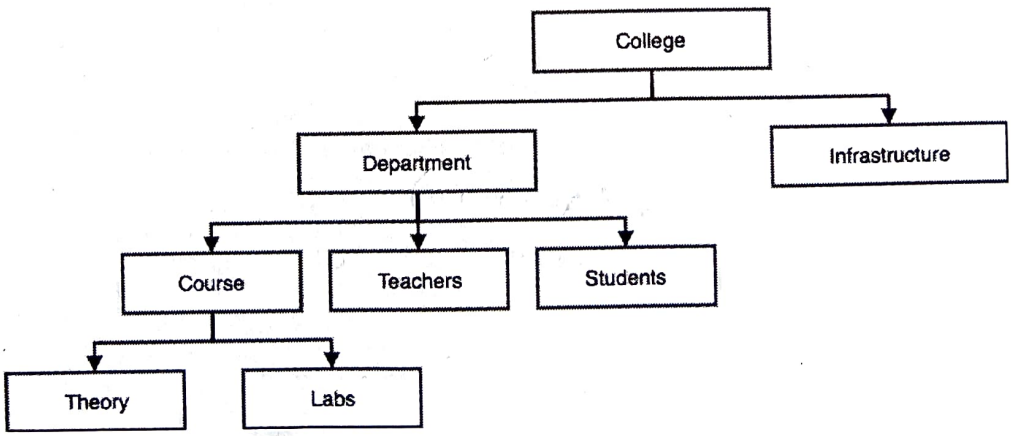
[5 * 3 = 15 Marks]

1. Compare and contrast AI, Machine Learning and Deep Learning with their scope and applications. [3] [CO01] [BTL 2]
2. Interpret the properties of the environment for the agent Stock Trading Algorithm and the PEAS representation for Spotify Recommendation System [An AI system that recommends songs based on listening habits and preferences]. [3] [CO01] [BTL 3]
3. Apply Cryptarithmetic for the following problem:

		C	R	O	S	S
+		R	O	A	D	S
=	D	A	N	G	E	R

[3] [CO02] [BTL 3]

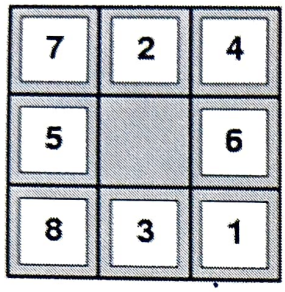
4.



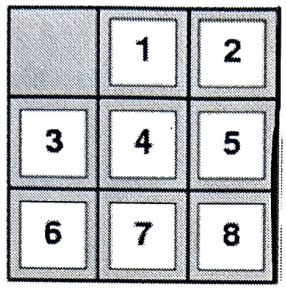
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Goal State: Students. Apply iterative deepening search to reach goal state and state the number of goal test needed to be performed. [3] [CO02] [BTL 3]

5. Calculate the heuristics $h_1(n)$ and $h_2(n)$ for the 8-puzzle number slider problem.
   $h_1(n)$ = number of misplaced tiles
   $h_2(n)$ = Manhattan distance to goal node



Start State

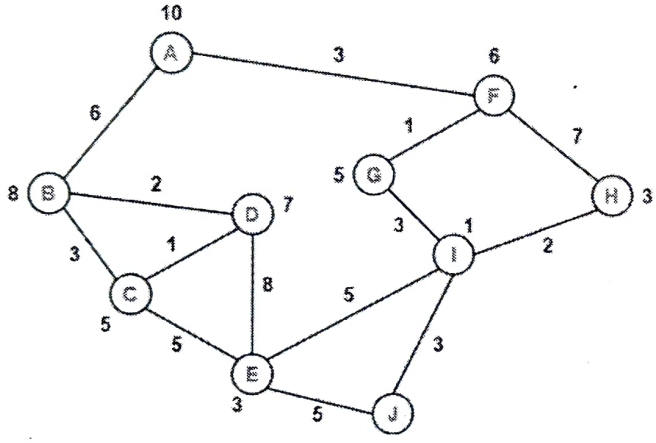


Goal State

[3] [CO01] [BTL 3]

PART B **[5 * 5 = 25 Marks]**

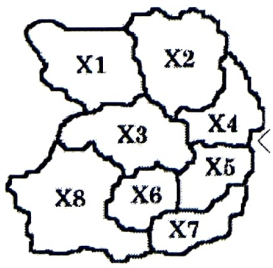
6. Solve the following using Heuristic-driven search and justify that it is a locally optimal search.



[5] [CO02] [BTL 3]

7. Analyse and explain in detail the working of the following as agents and their types:
   a. Thermostats
   b. Chess-Playing bot
   c. Pathfinder bot
   d. Stock trading bot
   e. DeepMind's AlphaGO [5] [CO01] [BTL 4]

8. Solve the following map coloring problem using any one technique of constraint satisfaction problem.



[5] [CO02] [BTL 3]

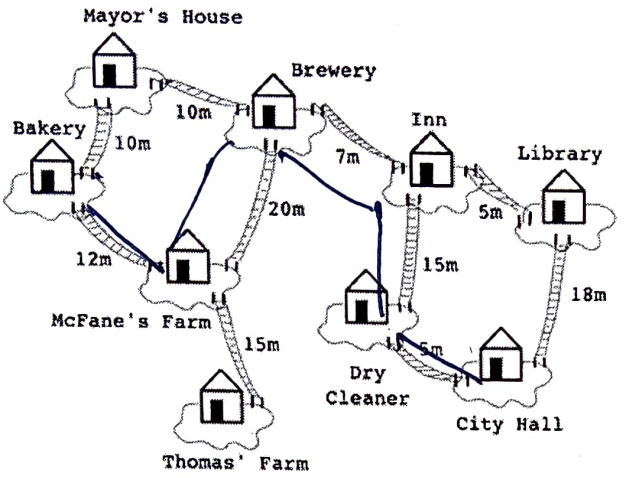
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9. Explain the Genetic Algorithm and justify why it is a local search optimisation algorithm. [5] [CO02] [BTL 2]

10. Compare and contrast BFS & DFS along with its COTS properties. [5] [CO02] [BTL 2]

PART C

11. Consider the following route map and the heuristics provided. Apply A* Search with initial state as Bakery and find the solution in a detailed stepwise manner. Prove the optimality of A* Search.



CITY HALL	0
INN	19
LIBRARY	16
DRY CLEANERS	2
BREWERY	5
MAYOR'S HOUSE	26
BAKERY	24
MCFANE'S FARM	17
THOMAS FARM	11

[10] [CO02] [BTL 3]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**14**	BTL 1	**3**
CO02	**36**	BTL 2	**10**
CO03	-	BTL 3	**32**
CO04	-	BTL 4	**5**
		BTL 5	
		BTL 6	

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